Neha Jha SQL/BI Developer

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PROFILE

Senior Database Engineer with 6+ years designing, engineering, and optimizing enterprise-grade data ecosystems across healthcare, financial services, an

d research environments.

Adept at collaborating with business leaders to translate analytical needs into scalable data architectures, operational automations, and governed reporting assets.

Proven ability to streamline data delivery by reducing end-to-end ETL runtimes by up to 60%, modernize legacy systems into cloud-aligned platforms, and build analytical processes capable of handling billions of records per day.

Collaborates with cross-functional stakeholders to shape analytical roadmaps, define data standards, and design scalable solutions aligned to regulatory, financial, and patient-care requirements.

Leads design of dimensional and relational data models, crafting high-performance schemas, star/snowflake structures, and enterprise data marts optimized for analytical workloads.

Engineers mission-critical SQL Server solutions using advanced T-SQL, indexing strategies, query tuning, and storage optimization to support high-volume, latency-sensitive applications.

Modernizes legacy database assets by redesigning schema structures, optimizing execution plans, and migrating workloads onto SQL Server 2019 and Azure SQL-based platforms.

Architects and orchestrates robust ETL/ELT pipelines using SSIS and Azure Data Factory, enabling fault-tolerant, metadata-driven data ingestion from APIs, files, structured, and semi-structured sources.

Builds high-throughput processing frameworks capable of transforming billions of daily records while enforcing auditability, lineage, and data governance controls.

Integrates Python-based automation and data processing routines to accelerate profiling, anomaly detection, and GPU-augmented analytics with accuracy improvements exceeding 70%.

Engineer enterprise reporting foundations using SSRS and Power BI, creating parameterized, drill-through, and executive dashboards that drive strategic and operational decision-making.

Develops and refines SSAS multidimensional cubes and tabular models, implementing hierarchies, partitions, KPIs, and MDX/DAX logic to support complex analytical queries at scale.

Establishes performance-engineering practices, including workload baselining, SSIS optimization, query refactoring, and server-level monitoring using Extended Events and Profiler.

Engineered high-volume data pipelines using Pandas and NumPy, optimizing data wrangling workflows and reducing end-to-end processing latency by up to 40%.

Built scalable data transformation frameworks leveraging vectorized Pandas operations and NumPy broadcasting, eliminating row-level loops and significantly improving throughput.

Applied SciPy statistical methods for anomaly detection, distribution fitting, and advanced data profiling to improve data quality and business rule validation.

Developed GPU-accelerated ETL workloads using NVIDIA CUDA and CuPy, speeding up complex aggregations and matrix computations by up to 10×.

Integrated Python-based ETL modules into enterprise orchestration systems (Airflow, ADF pipelines), enabling autonomously recoverable and fully traceable pipeline executions.

Designed reusable Python libraries for schema validation, data cleansing, partitioning, and metadata-driven transformations, ensuring cross-project consistency.

Built predictive and statistical data models using SciPy/NumPy and operationalized them into ETL flows for near–real-time decision-support systems.

Implemented robust data ingestion frameworks using Python to process structured, semi-structured, and unstructured data (CSV, Parquet, JSON, XML, and PDF).

Leveraged multiprocessing and async I/O to parallelize API and file ingestion tasks, achieving multi-fold performance gains over synchronous pipelines. Drives CI/CD automation using Azure DevOps to standardize deployment of SSIS packages, SQL objects, reporting assets, and environment configurations across development and production tiers.

Mentor engineers on database design, SSIS patterns, performance tuning, and reporting best practices, elevating team capability and standardizing engineering patterns.

Demonstrates a continuous-improvement mindset by adopting emerging data engineering practices, modernizing workflows, and integrating cloud-native advancements into established BI landscapes.

EDUCATION

Master of Science in Computer Science, University of Louisiana at Lafayette, Louisiana

SKILLS

Databases & Platforms: SQL Server (2025/2022/2019/2017), Azure SQL Database, Azure Synapse Analytics, PostgreSQL, Oracle 19c

Programming & Scripting: T-SQL, PL/SQL, Python (Pandas, NumPy, SciPy, CUDA), Shell Scripting, PowerShell

ETL/Data Integration: SSIS, Azure Data Factory, Informatica IICS, dbt, CDC, Change Tracking

BI & Analytics: SSRS, Power BI, Tableau, SSAS (Tabular/Multidimensional), Excel, DAX

Cloud & DevOps: Azure DevOps, Git, CI/CD Pipelines, Docker, Azure Pipelines, SQL Server Agent

Data Architecture: Star/Snowflake Schemas, Dimensional Modeling, Kimball Methodology, SCD Types 1-3

Specialized: GPU Computing (NVIDIA CUDA), Parallel Processing, Scientific Computing, API Integration

PROFESSIONAL EXPERIENCE

SQL/BI Developer

Serenity Healthcare | Irving, TX | June 2024 to Present

Responsibilities:

Architected enterprise-grade data warehouse supporting 50+ million patient records, implementing dimensional modeling with 15+ conformed dimensions and 8 fact tables, enabling real-time analytics for 200+ healthcare facilities

Engineered reusable, metadata-driven ETL framework reducing development time by 65% and onboarding new data sources from 2 weeks to 3 days through parameterized SSIS packages

Spearheaded migration of 120+ legacy ETL processes to modern SQL Server 2022 pipelines, achieving 40-60% reduction in runtime and 85% decrease in job failure rates through advanced error handling and checkpointing

Designed and implemented CDC framework processing 2+ billion transactional records daily with sub-5-minute latency, supporting real-time dashboards for executive decision-making

Optimized 200+ critical queries processing 10+ million records, reducing execution time by 75% through strategic indexing, partitioning (monthly/quarterly), and query rewriting techniques

Implemented temporal tables and system-versioned history for 30+ core tables, enabling time-travel queries and meeting regulatory compliance requirements for 7-year data retention

Conducted deep-dive performance audits across 50+ databases, implementing compression strategies and statistics updates that reduced storage by 40% and improved I/O throughput by 55%

Developed custom monitoring framework using DMVs and Extended Events, identifying and resolving 95% of performance bottlenecks before impacting production systems

Delivered 40+ enterprise Power BI dashboards and 60+ SSRS reports serving 500+ users, featuring drill-through capabilities, dynamic parameters, and mobile-responsive design

Built SSAS Tabular models with 100+ calculated measures using DAX, establishing unified semantic layer that reduced reporting inconsistencies by 90% across departments

Automated report distribution to 300+ stakeholders using Power BI Service subscriptions and SSRS data-driven subscriptions, eliminating 20+ hours/week of manual reporting

Established CI/CD pipelines using Azure DevOps for SSIS and SSAS deployments, reducing release cycle time from 2 days to 4 hours and achieving 99.8% deployment success rate

Implemented row-level security and role-based access control for 150+ users across Power BI and SSAS, ensuring HIPAA compliance and data protection

Created comprehensive data lineage documentation and data dictionaries for 200+ tables, improving data literacy and reducing onboarding time for new developers by 50%

Mentored team of 5 junior developers on T-SQL optimization, SSIS best practices, and dimensional modeling, increasing team productivity by 35%

Environment: SQL Server 2022/2019, Azure Data Factory, Azure Synapse Analytics, T-SQL, Oracle 23c, SSIS, SSAS (Tabular), SSRS, Power BI, Python, C#, Windows Server, Visual Studio, Azure DevOps, Git

RESEARCH DATA ENGINEER & PYTHON TEACHING ASSISTANT

University of Louisiana at Lafayette | Lafayette, LA | August 2022 to May 2024

Responsibilities:

Optimized GPU-accelerated weather prediction models using Python and NVIDIA CUDA, enabling parallel processing of 30+ concurrent model instances with 73.6% improvement in forecasting accuracy

Engineered high-throughput data pipelines processing 5+ terabytes of atmospheric data daily from 500+ weather stations using Python multiprocessing, achieving sub-minute latency for time-critical forecasting

Designed multi-petabyte meteorological data warehouse with optimized star schema, supporting complex temporal queries across 20+ years of historical weather data for climatological research

Developed real-time streaming architecture using Python asyncio processing 10,000+ observations/minute from NOAA, ECMWF, and GFS APIs with millisecond-precision timestamping

Created 50+ stored procedures with advanced window functions and CTEs, processing billions of weather records for statistical analysis and model validation with 10x performance improvement

Implemented automated data quality framework detecting anomalies in 99.7% of data inconsistencies across multi-source meteorological datasets, ensuring model input reliability

Built SQL-based model validation pipelines computing forecast skill scores and bias corrections for 15+ atmospheric variables, enabling automated performance assessment

Instructed 120+ undergraduate students in advanced Python programming, covering NumPy, SciPy, Pandas, parallel processing, CUDA programming, and database integration for scientific computing

Developed hands-on curriculum with 20+ projects focused on atmospheric data analysis, achieving 95% student success rate in applying GPU optimization techniques

Environment: SQL Server 2019, Windows Server 2019, Visual Studio, T-SQL, SSIS, Python 3.x, NVIDIA CUDA, NumPy, SciPy, Pandas, Multiprocessing, Threading, Asyncio, Java, REST APIs, Git, Linux/Unix, Docker, Jupyter Notebooks

SSIS/SSRS Developer

Cedar Gate Technologies | Kathmandu, Nepal | April 2021 to July 2022

Responsibilities:

Designed and implemented dimensional data models with 25+ dimensions and 10+ fact tables, supporting healthcare claims processing for 2+ million members

Developed 80+ SSIS packages with advanced transformations (Lookup, SCD, Conditional Split), achieving 99.5% data accuracy through comprehensive validation and error handling

Built metadata-driven ETL framework reducing package development time by 50% through dynamic configurations, parameterization, and reusable components

Implemented SCD Type 2 for 15+ dimension tables, maintaining complete historical tracking of member demographics and provider information

Designed and deployed 8 SSAS multidimensional cubes with 40+ measures and 20+ calculated members using MDX, serving 100+ business analysts

Created 35+ SSRS reports featuring drill-through, matrix, and tabular layouts integrated with SSAS cubes for executive healthcare analytics

Automated 50+ recurring data processing jobs using SQL Server Agent, reducing manual intervention by 90% and ensuring 99% on-time execution

Optimized query performance through indexing strategies and execution plan analysis, reducing average query time from 45 seconds to 6 seconds (87% improvement)

Conducted data profiling on 100+ source tables, identifying and resolving data quality issues improving downstream analytics reliability by 95%

Environment: SQL Server 2017, Windows Server 2019, Visual Studio, T-SQL, SSIS, SSRS, SSAS (Multidimensional), C#, TFS, Control-M

Junior SQL Developer/ Data Engineer

Bytecode Technologies | Kathmandu, Nepal | October 2018 to March 2021

Responsibilities:

Built 60+ SSIS packages with robust error handling and logging mechanisms, processing 500K+ records daily from heterogeneous sources

Developed 100+ database objects (tables, stored procedures, views, functions) supporting loan processing applications for 50K+ customers

Created dynamic ETL solutions with parameterization enabling cross-platform data movement and reducing maintenance overhead by 40%

Designed and delivered 45+ SSRS reports (tabular, matrix, drill-down) providing operational metrics for loan processing workflows

Built SSAS cubes with star and snowflake schemas, enabling multidimensional analysis for business performance monitoring

Automated report distribution using SQL Server Agent, serving 75+ stakeholders with daily, weekly, and monthly KPI dashboards

Environment: SQL Server 2016, Visual Studio, T-SQL, Azure SQL, SSIS, SSRS, SSAS, TFS, Windows Server 2016